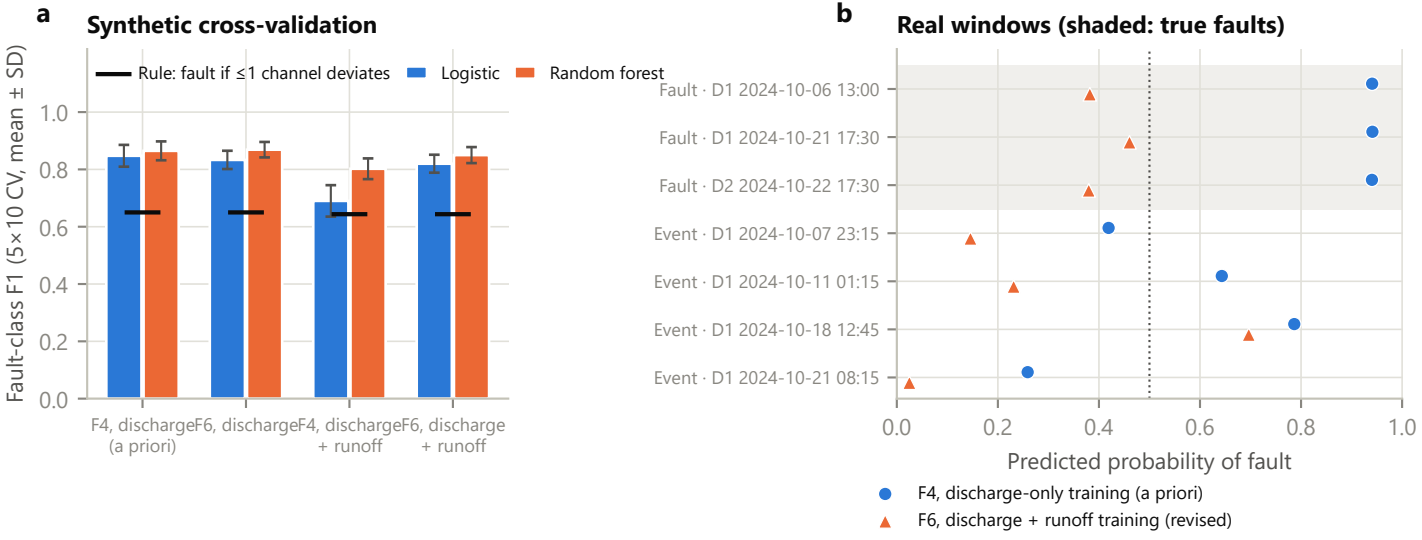
a
Synthetic cross-validation
Rule: fault if ≤1 channel deviates
Logistic
Random forest
Fault-class F1 (5×10 CV, mean ± SD)
1.0
0.8
0.6
0.4
0.2
0.0
F4, discharge (a priori)
F6, discharge
F4, discharge + runoff
F6, discharge + runoff
b
Real windows (shaded: true faults)
Fault · D1 2024-10-06 13:00
Fault · D1 2024-10-21 17:30
Fault · D2 2024-10-22 17:30
Event · D1 2024-10-07 23:15
Event · D1 2024-10-11 01:15
Event · D1 2024-10-18 12:45
Event · D1 2024-10-21 08:15
0.0
0.2
0.4
0.6
0.8
1.0
Predicted probability of fault
F4, discharge-only training (a priori)
F6, discharge + runoff training (revised)